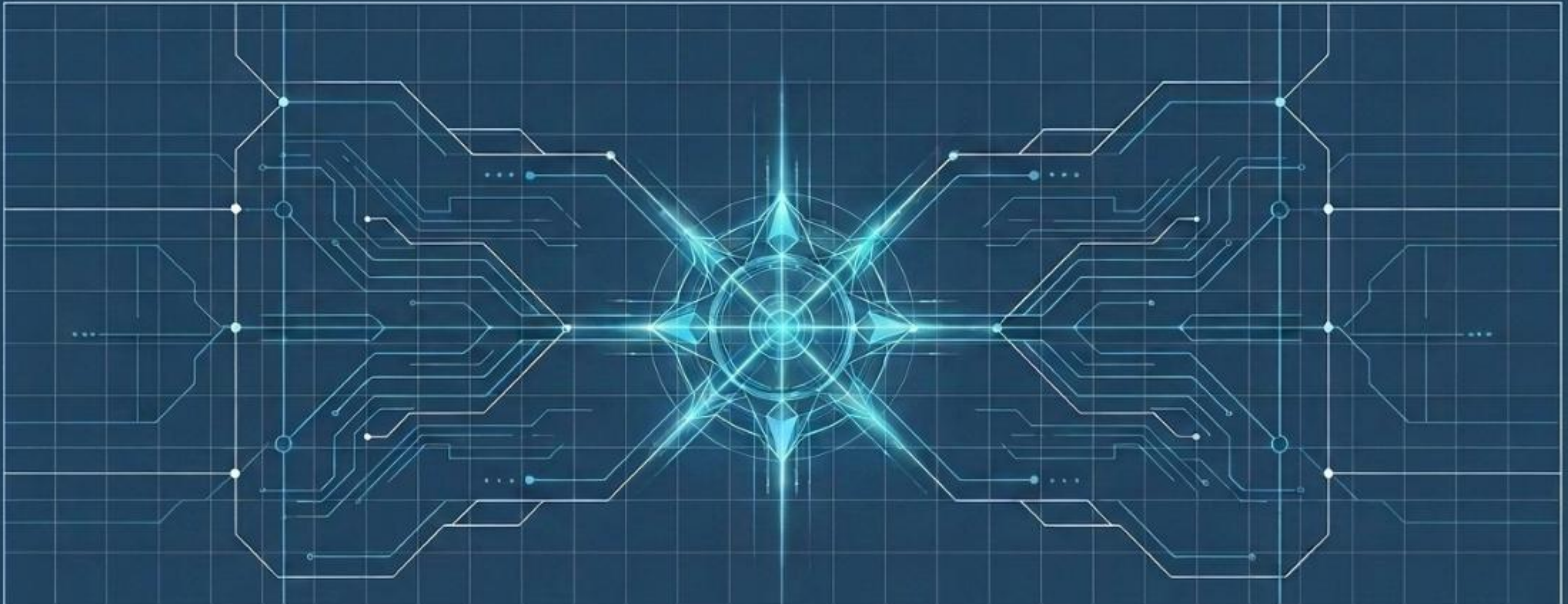




COURSE:
ETHICAL HACKING (ITEC306)

ETHICAL HACKING

THE 39-SECOND REALITY

39 SECONDS

INTERVAL_CYBERATTACK

\$ BILLIONS

ANNUAL_FINANCIAL_LOSS_EST



GLOBAL FREQUENCY

Every 39 seconds, a cyberattack occurs globally.



RATE: 15,384 ATTEMPTS/HOUR

ECONOMIC IMPACT

Billions of dollars are lost annually to data breaches, ransomware, and system intrusions.



COST: >\$1 TRILLION/YEAR

SOCIETAL FOUNDATION

Information security is no longer an optional IT function; it is the foundation of modern society.



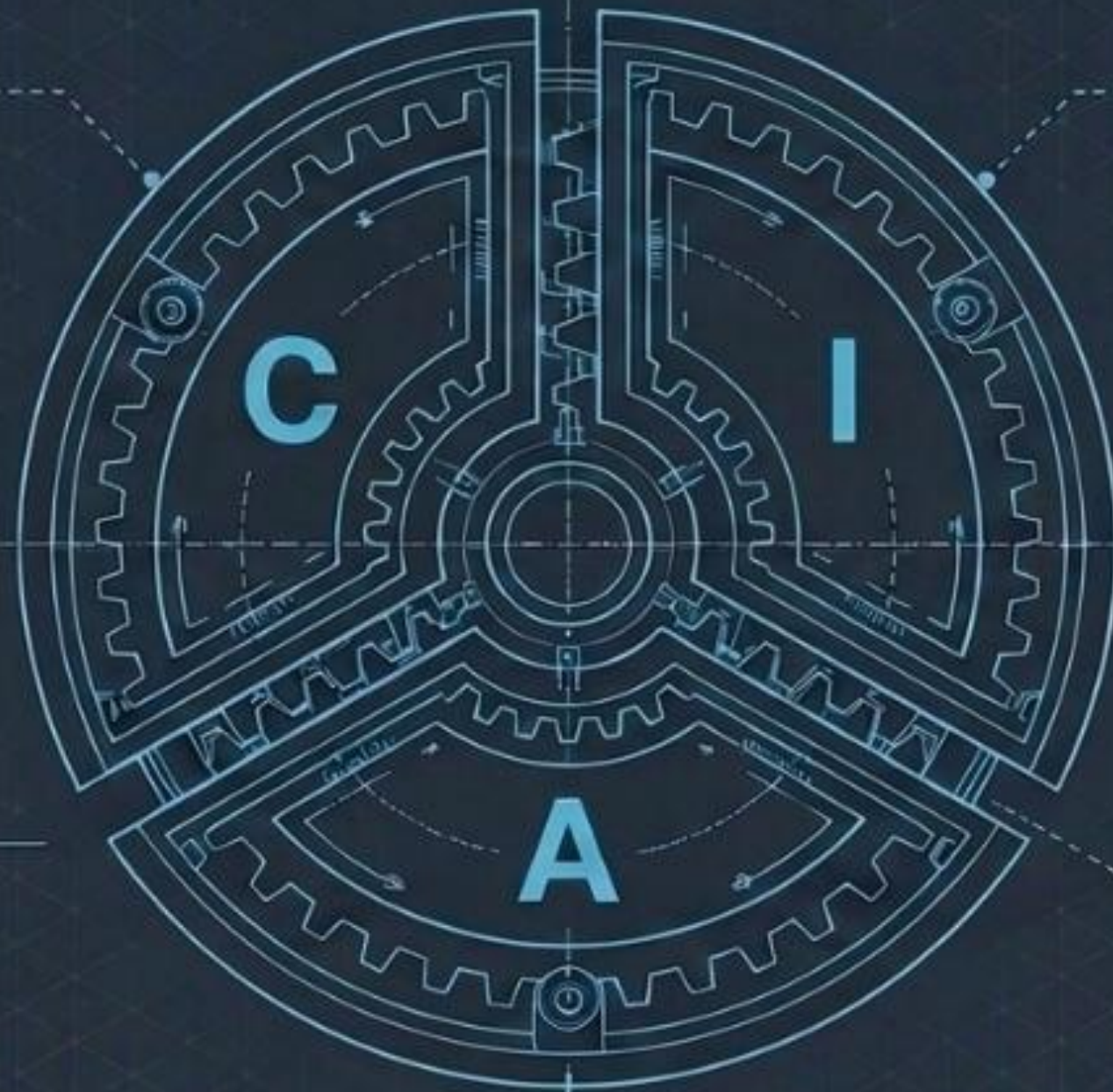
CRITICAL INFRASTRUCTURE DEFENSE

The Foundation: The CIA Triad

Confidentiality: Ensuring that information is accessible only to authorized individuals. It prevents unauthorized access to data, keeping it private and protected.

 **Control:** Encryption, MFA, access control.

 **Threat:** Eavesdropping, data breaches, credential theft.



Integrity: Ensuring that information is trustworthy and has not been altered, deleted, or modified by unauthorized people.

 **Control:** Hashing (SHA-256), digital signatures.

 **Threat:** Man-in-the-middle, malware, data interference.

Availability: Ensuring that information and systems are accessible to authorized users when needed.

 **Control:** Redundant servers, backups, DDoS protection.

 **Threat:** DoS attacks, ransomware, hardware failure.

UNDERSTANDING DEFENSE STRATEGIES: REACTIVE VS. PROACTIVE

REACTIVE DEFENSE



TRADITIONAL CONTROLS: Firewalls & Antivirus.

These are like walls and locks that wait for an attack to happen. They are necessary, but they only protect against known threats and cannot identify hidden weaknesses. You cannot fix vulnerabilities you do not know exist.

PROACTIVE DEFENSE



ETHICAL HACKING: Actively identifying weaknesses.

Instead of waiting, ethical hacking thinks like an attacker to find and fix vulnerabilities before bad guys exploit them. It provides measurable, objective evidence of security posture before a breach occurs. It's about staying one step ahead.



PROJECT: CYBERSECURITY FOUNDATIONS

DOCUMENT ID: ETH-HACK-PS-03

ENGINEER: CYBERSEC CONSULTANTS

WHAT IS ETHICAL HACKING?

The practice of intentionally probing computer systems, networks, and applications to find security vulnerabilities that a malicious hacker could exploit.

The difference between a criminal hacker and an ethical hacker isn't the tools they use—it's **intention, permission, and authorization.**



PHO-HIT, CYBERSECURITY FOUNDATION'S
DOCUMENT ID: ETH-HACK-LC-05
BHCAREER, CYBERSEC CONSULTANTS

THE THREE PILLARS OF ETHICAL HACKING LEGALITY



1. WRITTEN AUTHORIZATION

You must have a signed contract and explicit written permission from the owner of the infrastructure you are testing. Without this, accessing a system is a crime.



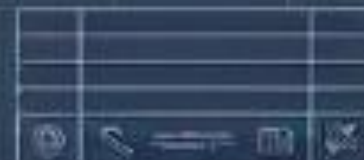
2. SCOPE & BOUNDARIES

You must stay within the boundaries defined in the contract. If a company says, "Test our website," you do not test the email server.



3. ETHICS & CODE OF CONDUCT

You must adhere to a code of conduct. This includes maintaining confidentiality, not stealing data, and not causing damage to the system beyond what is necessary to prove a vulnerability.



THE LOCKSMITH PREMISE

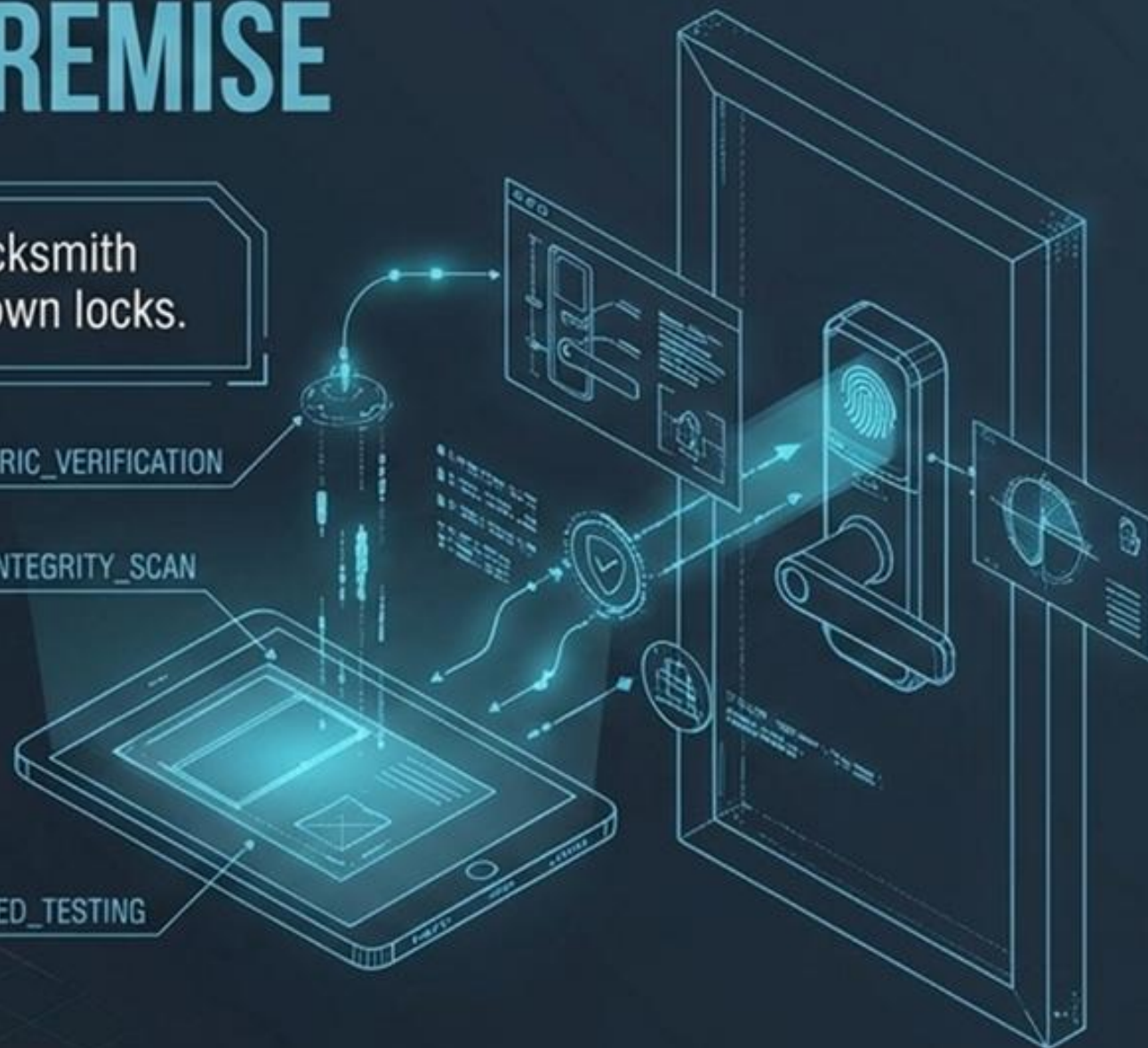
Think of an ethical hacker as a locksmith hired by a homeowner to test their own locks.



BIOMETRIC_VERIFICATION

SYSTEM_INTEGRITY_SCAN

AUTHORIZED_TESTING



EXPLICIT PERMISSION

Authorized access to test.



DEFINED SCOPE

Specific boundaries set.



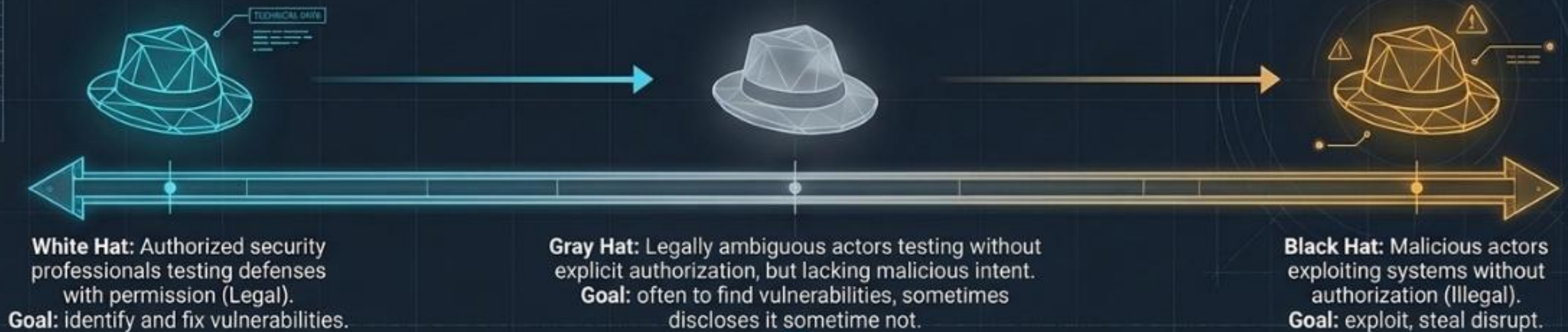
GOAL OF MAKING THE HOME SAFER

Improve overall security.



Behind every secure system stands someone who understands exactly how attackers think.

The Threat Landscape: Hacker Classifications





Anonymous Philippines

Greetings Philippines! We are Anonymous.

The Constitution so asserts that "Sovereignty resides in the people and all government authority emanates from them."
One of the processes by which people exercise their sovereignty is through voting in an election- where people choose the candidates who will best represent them, who will serve them under the principle that "Public office is a public trust."
But what happens when the electoral process is so mired with questions and controversies?
Can the government still guarantee that the sovereignty of the people is upheld?

We request the implementation of the security features on the PCOS machines.

Commission on Elections, we are watching!

We are Anonymous,
We are legion,
We do not forgive,
We do not forget.

EXPECT US!

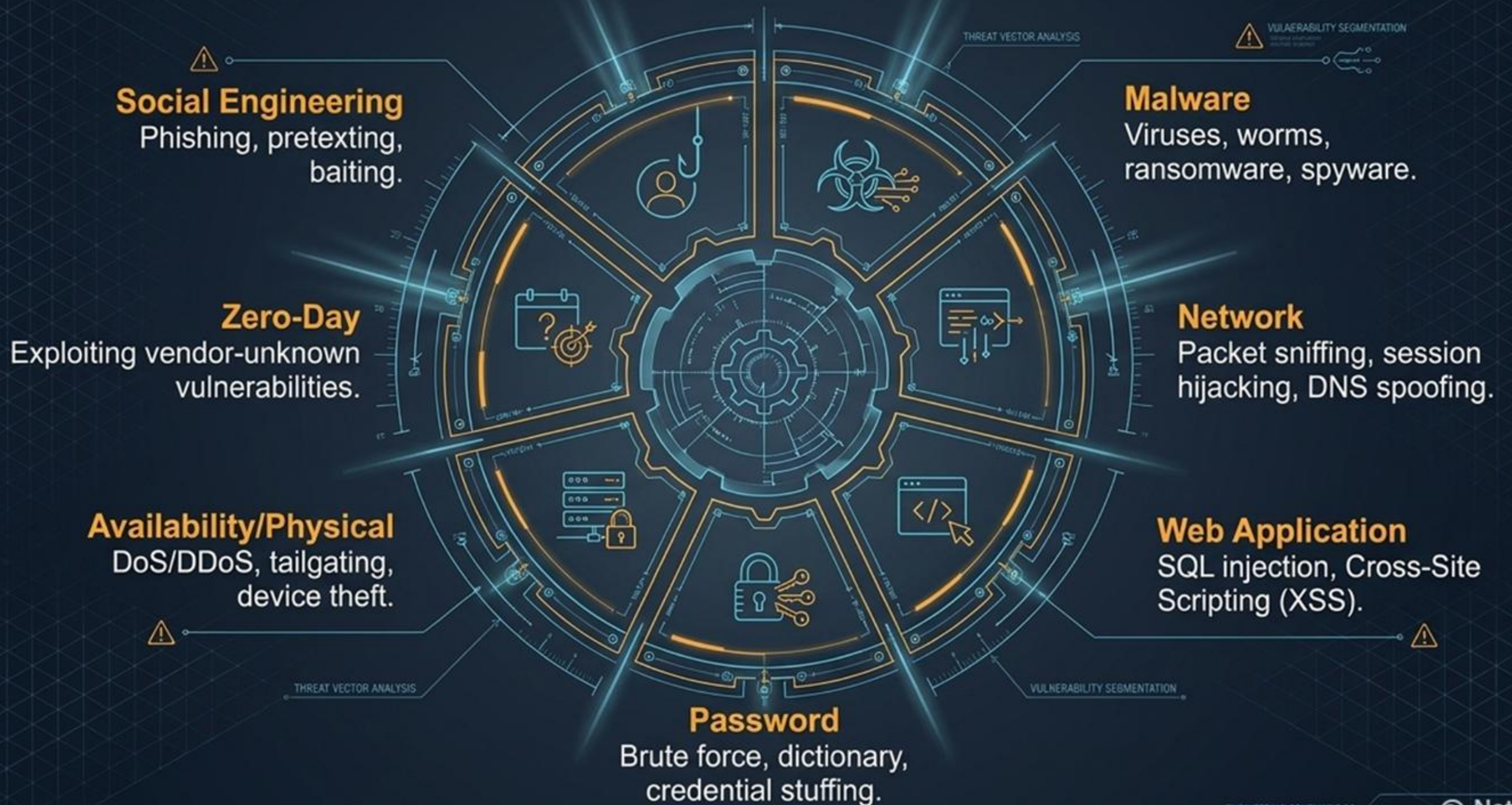
Are you a concerned citizen? Mail us at info@anonph.net

-a1farious | ./kh4lifax | sec.anonops

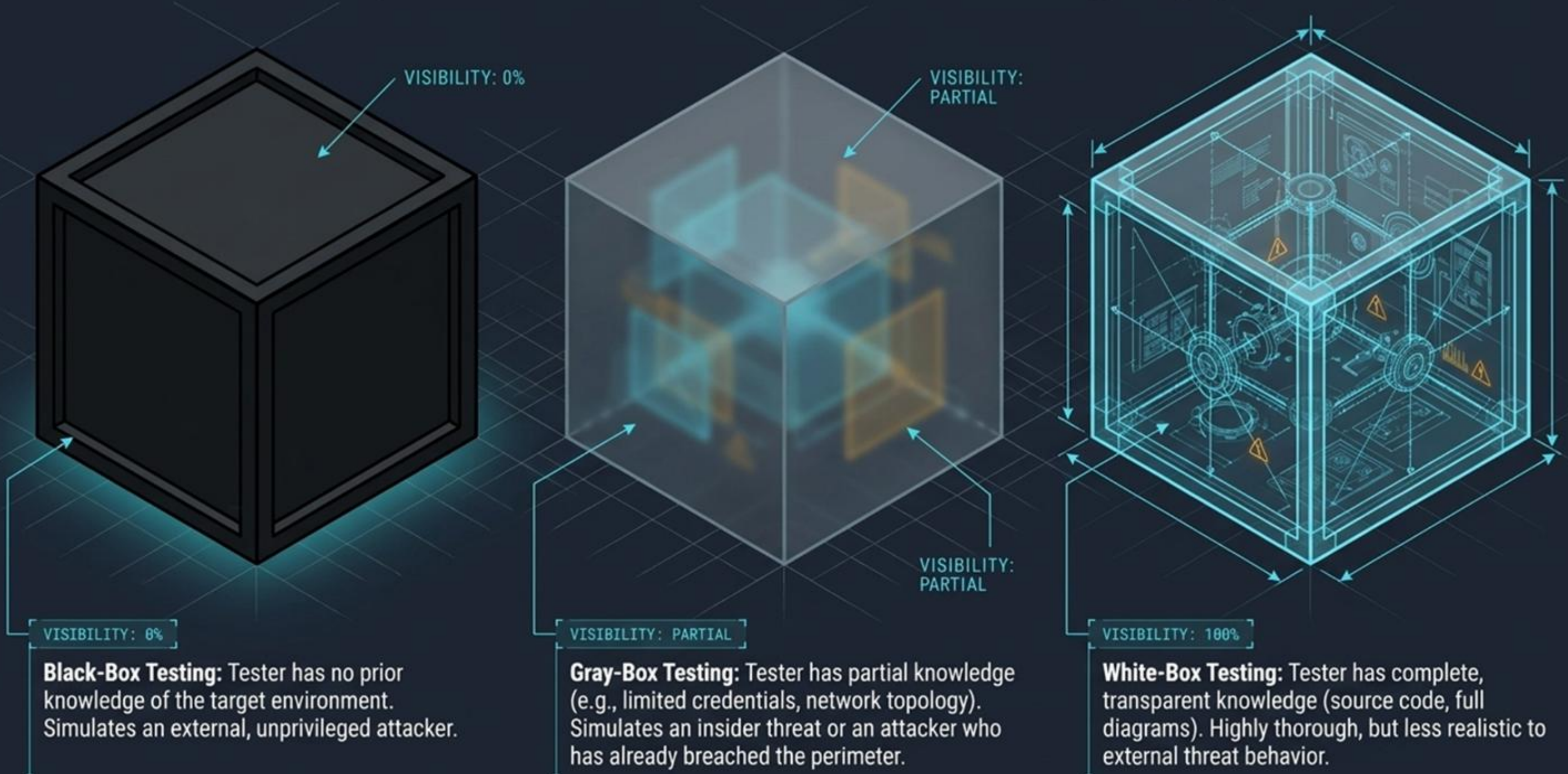
Anonymous Philippines - Phantom Hackers.PH - Philippine Cyber Army - PrivateX

#pr.1stair | busabos | POWERFUL | -manilacoy | s4v4nt.r00t | -invictus | musarva rider | Blood Sec Hackers

The Attacker's Toolkit



Scope of Vision: Box Testing Types



The legal shield: Pre-engagement documentation



Key 1: Statement of Work (SOW)

The legally binding commercial agreement defining deliverables, timeline, and cost.

Key 2: Non-Disclosure Agreement (NDA)

Protects client infrastructure details and prevents third-party vulnerability disclosure.

Key 3: Rules of Engagement (ROE)

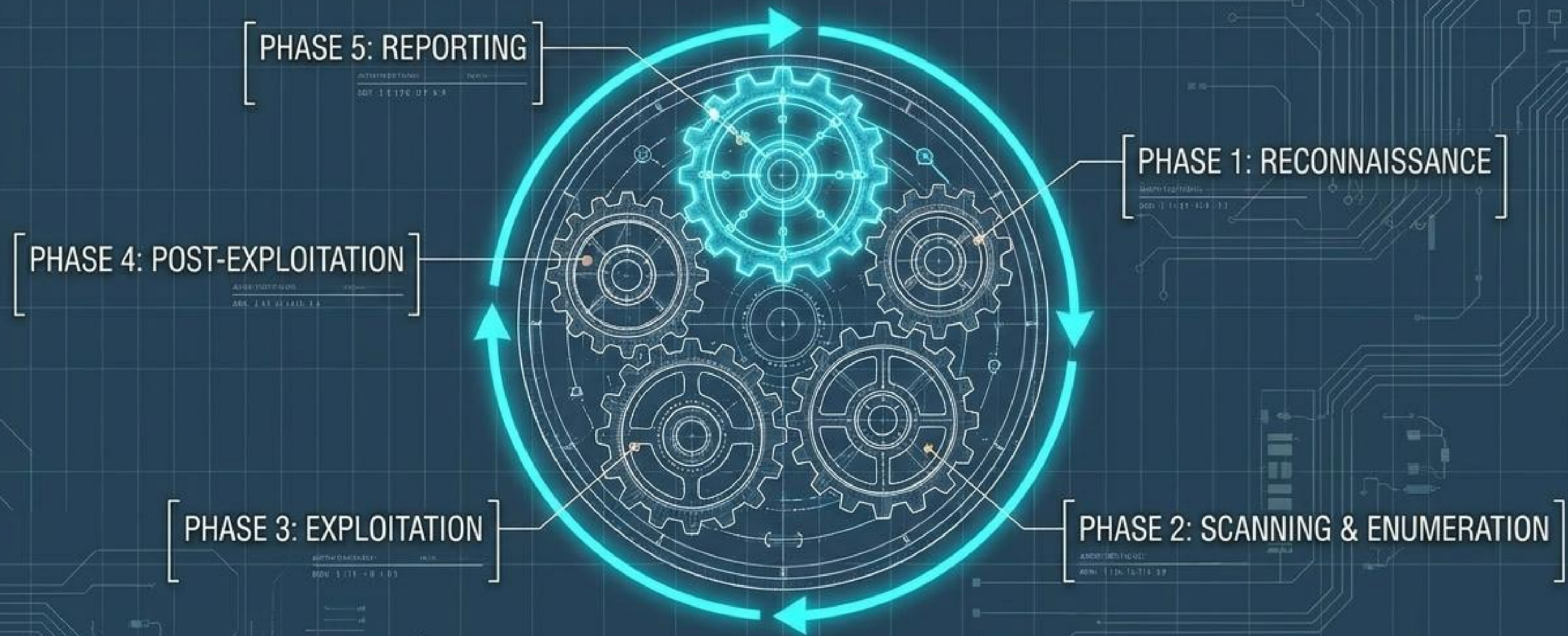
Defines in-scope/out-of-scope systems, testing windows, prohibited techniques, and data handling.

THE 5-PHASE ETHICAL HACKING PROCESS



PWD/CRT: CYBERSECURITY FOUNDATIONS
DOCUMENT ID: ETH-HACK-DC-05
ENGINEER: CYBERSEC CONSULTANTS

THE 5-PHASE PENETRATION TESTING LIFECYCLE



The lifecycle ensures thoroughness, legal compliance, and professional, repeatable deliverables.



PROF. CYBERSECURITY FOUNDATIONS
DOCUMENTED ETH-HACK-LC-05
ENGINEER CYBERSEC CONSULTANTS

PHASE 1: RECONNAISSANCE

The Intelligence-Gathering Phase

OBJECTIVE

Collect maximum information about the target environment without raising alarms.

METHODOLOGY

Open Source Intelligence (OSINT), WHOIS lookups, DNS enumeration, Google Dorking.

TOOLKIT EXAMPLES

Maltego, theHarvester, Shodan, Censys.



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DOCUMENT ID: ETH-HACK-RC-06
ENGINEER: CYBERSEC CONSULTANTS

PASSIVE VS. ACTIVE RECONNAISSANCE

[PASSIVE RECONNAISSANCE]



ZERO DIRECT INTERACTION

Querying public databases, search engines, and social media. The target has no way of knowing they are being investigated.

[ACTIVE RECONNAISSANCE]



DIRECT SYSTEM INTERACTION

Ping sweeps, port scanning, and banner grabbing. Packets are sent directly to the target infrastructure.

LEGAL WARNING: Active reconnaissance sends packets to target systems. Unauthorized port scans can trigger IDS/IPS and violate laws. Always confirm scope before scanning.



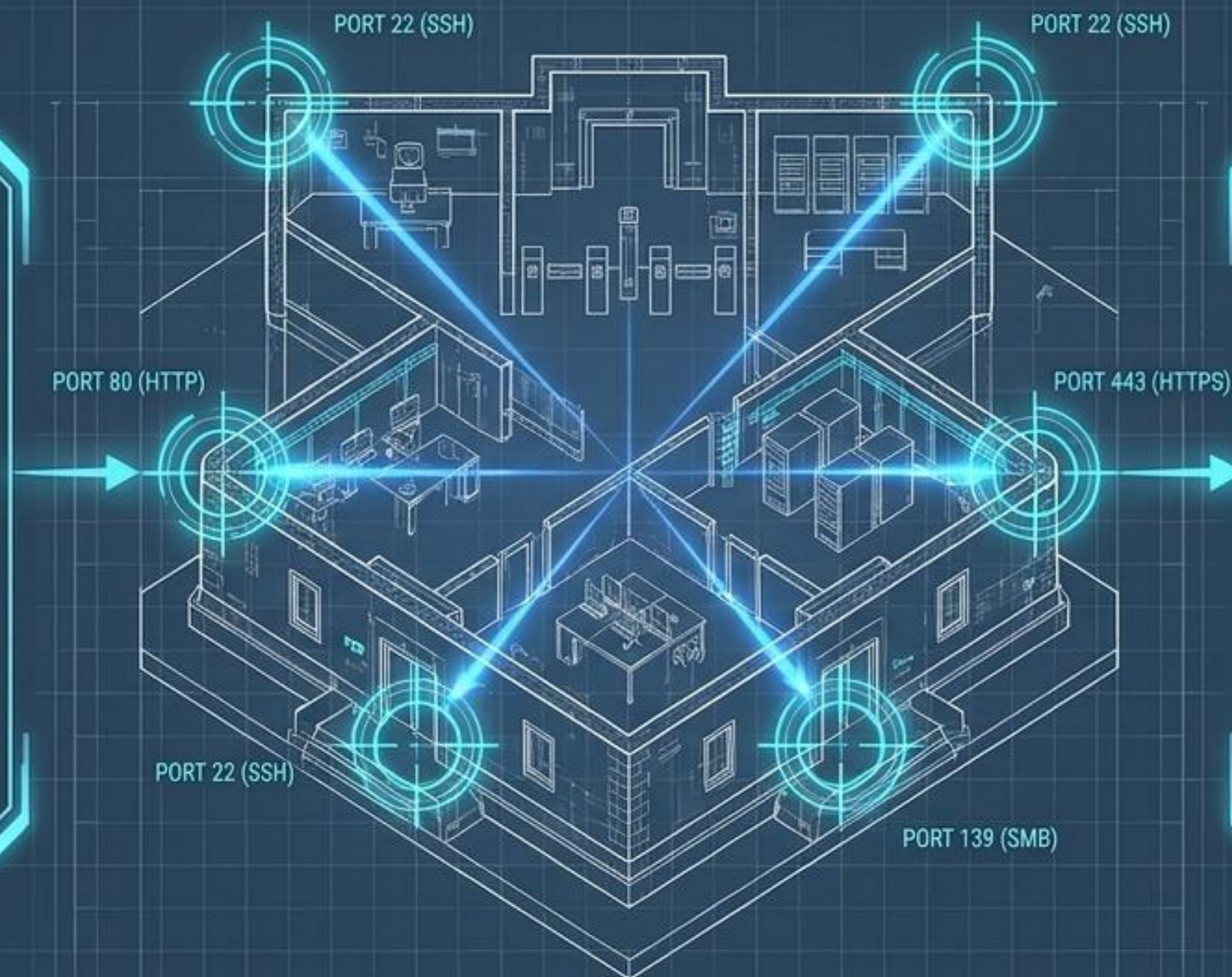
PWO / CRT. CYBERSECURITY FOUNDATIONS
DOCUMENT ID ETH-HACK-RC-07
ENGINEER CYBERSEC CONSULTANTS

PHASE 2: SCANNING & ENUMERATION

NETWORK SCANNING

Transforming the broad intelligence map into a targeted inventory. Identifying live hosts, open TCP/UDP ports, and OS versions.

Key Tools:
Nmap, Masscan



DEEP ENUMERATION

Systematically querying discovered services to extract usernames, network shares, routing tables, and banner data.

Key Targets:
SMB, SNMP, LDAP,
Web Directories

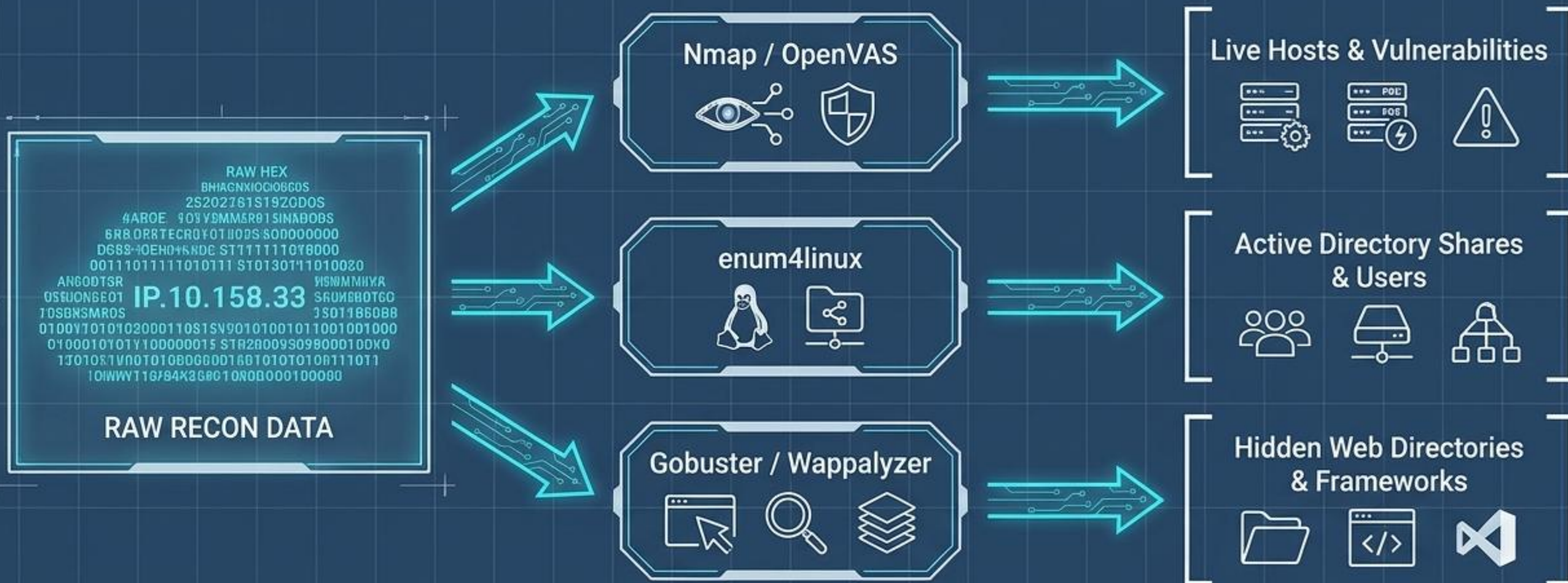


PROJECT: CYBERSECURITY FOUNDATIONS

DOCUMENT NO: ETH-HACK-RC-08

ENGINEER: CYBERSEC CONSULTANTS

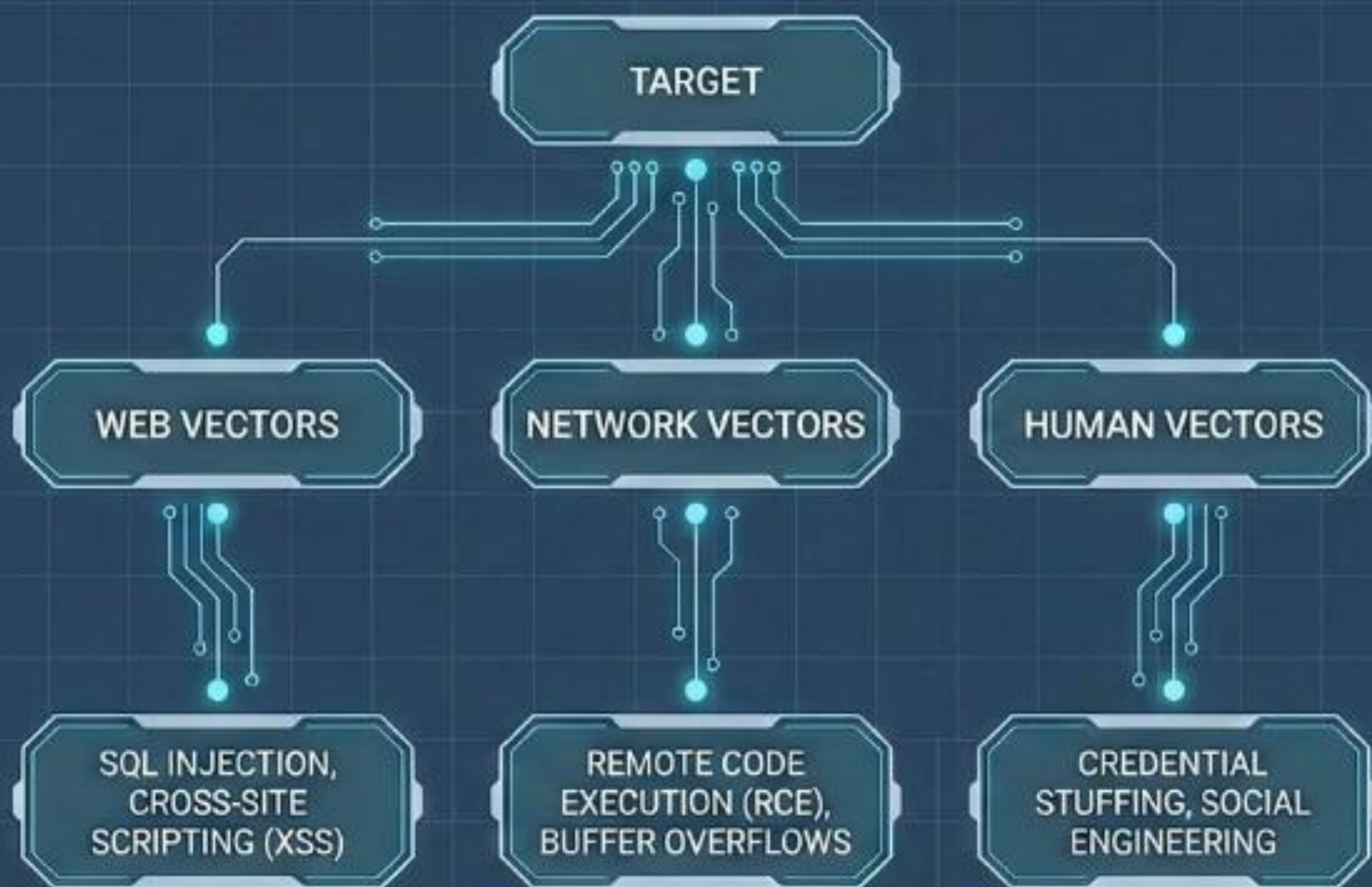
[THE ENUMERATION TOOLKIT]



Enumeration requires precision. Automated tools are only as effective as the tester's ability to analyze their output.

PHASE 3: EXPLOITATION

The Critical Strike Phase

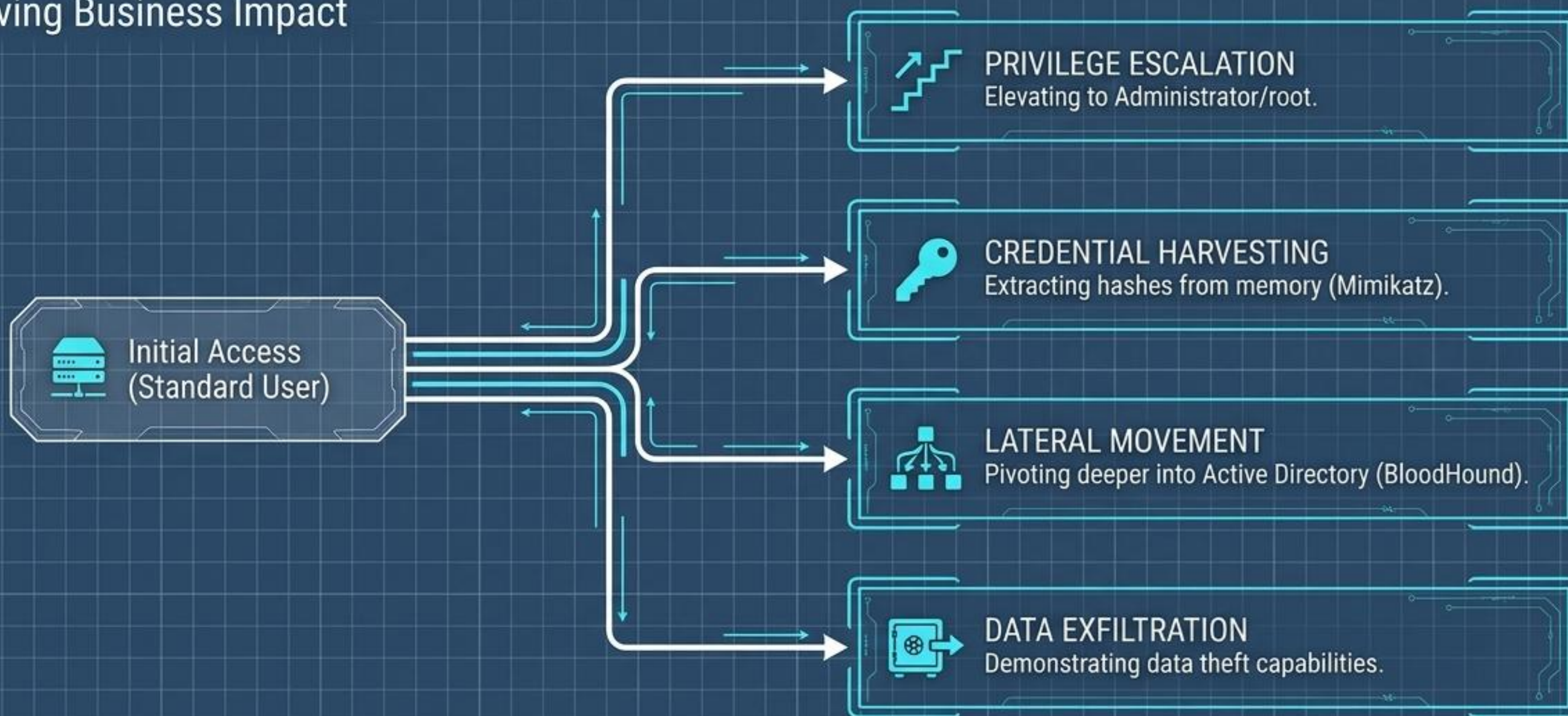


EXPLOITATION PHASE:

the hacker is attempting to gain an unauthorized access by bypassing security controls and exploiting identified vulnerabilities gathered from scanning phase.

PHASE 4: POST-EXPLOITATION

Proving Business Impact



Initial access is just the beginning. Post-exploitation demonstrates what an Advanced Persistent Threat (APT) could realistically accomplish.

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DOCUMENT NO: ETH-HACK-RC-11

ENGINEER: CYBERSEC CONSULTANTS

PHASE 5: REPORTING

The Only Deliverable That Matters

1. EXECUTIVE SUMMARY:
Plain language business impact
and overall risk rating.

2. SCOPE & METHODOLOGY:
Systems tested, testing
approach, and dates.



3. FINDINGS:
Vulnerability descriptions,
CVSS scores, and evidence.

4. REMEDIATION:
Prioritized, actionable steps
to fix each finding.

5. APPENDICES:
Raw tool output,
proof-of-concept code,
and network diagrams.

A technically brilliant penetration test is entirely worthless if the findings cannot be communicated clearly to executive leadership.

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DOCUMENT NO: ETH-HACK-RC-12

ENGINEER: CYBERSEC CONSULTANTS

The Ethical Hacker's Arsenal

Phase 1 (Recon)

Maltego

theHarvester

Shodan

OSINT frameworks

Phase 2 (Scanning)

Nmap
(Host/Port discovery)

Nessus/OpenVAS
(Vulnerability scanning)

Gobuster

Phase 3 (Exploitation)

Metasploit Framework

Burp Suite

SQLMap

Phase 4 (Post-Exploitation)

Mimikatz
(Credential extraction)

BloodHound
(Active Directory paths)

Cobalt Strike

Phase 5 (Reporting)

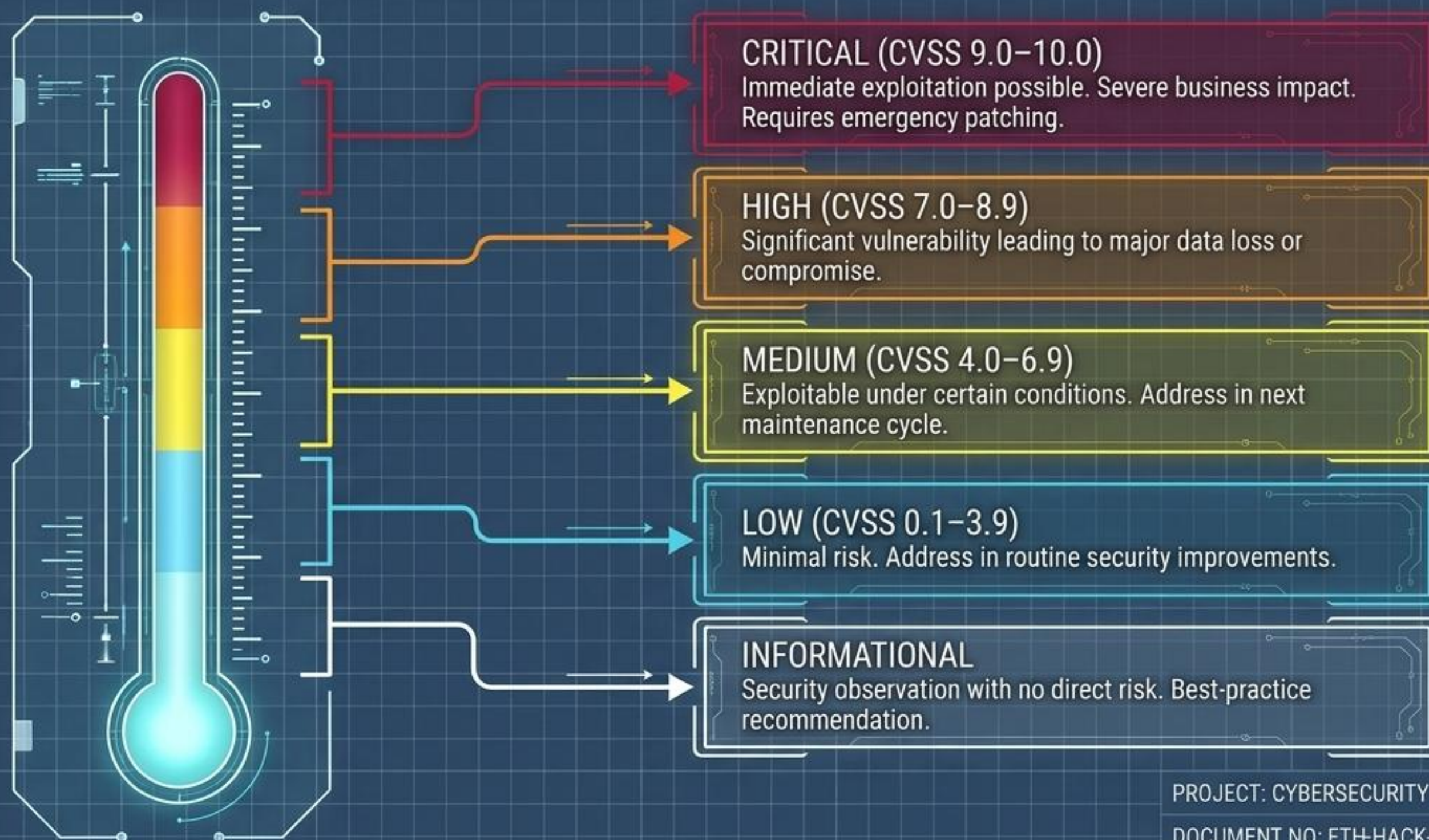
Dradis

Serpico

MS Word

An ethical hacker uses the exact same toolkit as a malicious actor, but applies the EC-Council CEH methodology to ensure structured, safe, and remediable outcomes.

VULNERABILITY RATING SCALES



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DOCUMENT NO: ETH-HACK-RC-13

ENGINEER: CYBERSEC CONSULTANTS

The Tactical Battlefield



RED TEAM The Offense.

Simulates real-world cyber attacks to expose vulnerabilities before they can be exploited.

Two sides of the same coin,
fighting a simulated war to
achieve true security.



BLUE TEAM The Defense.

Provides continuous security monitoring, threat detection, and incident response across the organization.

Objectives & Responsibilities

Objective

Simulate real-world cyber attacks to identify vulnerabilities and security gaps.

Responsibilities

Test detection and incident response capabilities.

Provide recommendations on how to remediate vulnerabilities and patch gaps.

Objective

Provide security monitoring and detection across systems, networks, and applications.

Responsibilities

Investigate, contain, and respond to live cyber security incidents.

Secure the perimeter and internal network.

Tactical Toolkits & Techniques



Kali Linux (Primary OS environment)

Network exploitation tools & Penetration testing methodology

Password cracking & Privilege escalation scripts

Intercepting proxies & C2 (Command & Control) frameworks



SIEM (Security Information and Event Management)

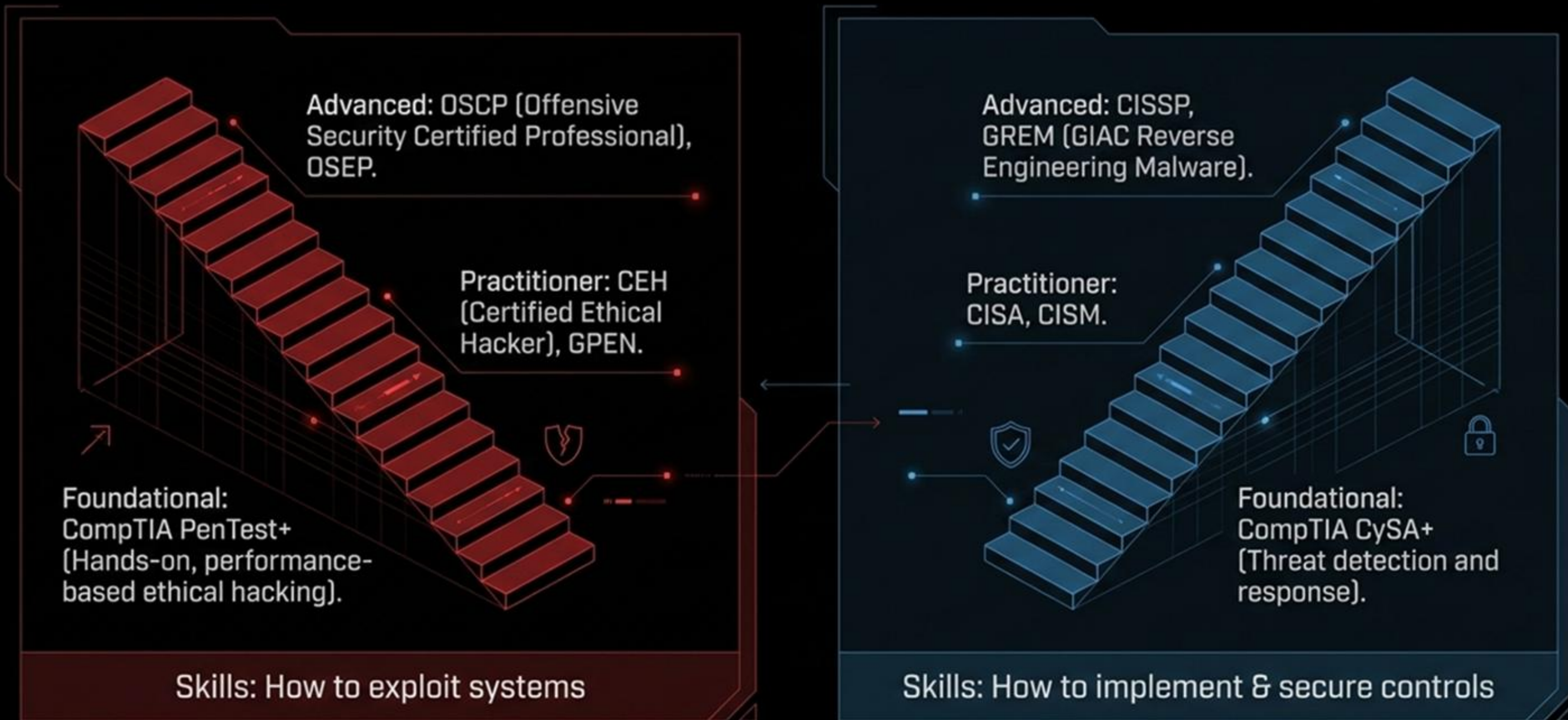
IDS / IPS (Intrusion Detection/Prevention Systems)

EDR (Endpoint Detection and Response)

Wireshark, Malware analysis & Reverse engineering tools

Forensic analysis & IAM tools

The Certification Ladder



Differing Arsenals: Tools and Tactical Skills

KEY SKILLS	Knowledge of how to exploit systems. Network Scanning. Penetration Testing.
TOOLS & TECHNIQUES	Kali Linux, password cracking tools, intercepting proxies, network exploitation tools, privilege escalation scripts, and C2 frameworks.

KEY SKILLS	Knowledge of how to implement and secure security controls. Implementing Detection Systems.
TOOLS & TECHNIQUES	IDS, IPS, EDR, SIEM, malware analysis, reverse engineering tools, Wireshark, IAM tools, and cloud security tools.

Summarize

